

Data Analysis Report: Attentional spatial bias in two image tasks

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Report

Final report for eye tracking course.

Source code and data: <https://github.com/jnagasawa/ET-report>

Chapter 1

Background

A robust leftward bias in early visual exploration has been documented across diverse tasks, including free scene viewing, line bisection, face perception, and two-alternative forced choice paradigms (Foulsham et al. 2013; Ossandón et al. 2014). In speakers of left-to-right (LTR) languages, the proportion of initial fixations landing on the left side of a display typically falls between 60% and 93%, depending on the task and stimulus material (Ehinger et al. 2026). Because this preference operates independently of image content, it can substantially reduce the sensitivity of paradigms designed to measure content-driven attentional biases.

The origin of the leftward bias remains debated. A dominant account links it to the right-hemisphere lateralization of the attentional network, which preferentially represents the left hemifield (Corbetta and Shulman 2002). This explanation does not, however, readily account for cross-linguistic variation: speakers of Arabic and Hebrew show a weaker or reversed preference in line bisection and related tasks (Chokron and Imbert 1993; Rashidi-Ranjbar et al. 2014), suggesting that habitual reading direction modulates the baseline spatial asymmetry.

Afsari et al. (2016) examined this modulation directly using a priming paradigm. Bilingual participants read either RTL or LTR text passages before freely viewing photographs. Native RTL speakers (Arabic, Urdu, or Persian) showed an early rightward shift in fixation after reading RTL primes, but a leftward shift after LTR primes. Native LTR speakers who had acquired an RTL language later in life did not show comparable flexibility. These results indicate that reading direction influences spatial scanning dynamically, and that the effect depends on habit formation during native language acquisition rather than current language use alone.

Complementary evidence concerns the consequences of spatial biases for experimental design. Ehinger et al. (2026) measured first-fixation preferences in a concurrent two-stimulus paradigm with stimulus pairs placed at six angular positions relative to a central fixation cross. In the standard horizontal (9 o'clock / 3 o'clock) arrangement, German-speaking participants directed their first saccade to the left stimulus on approximately 80% of trials, with a comparable upward bias in the vertical arrangement. A diagonal placement at 1 o'clock and 7 o'clock yielded no net spatial preference, because the leftward and upward biases cancelled. Simulations based on a Bradley-Terry model showed that eliminating the spatial bias can increase statistical power by 30% or more for small

attentional effects.

1.1 Motivation

The literature on spatial scanning biases has been developed almost exclusively with LTR-reading populations, and spatial biases are routinely counterbalanced rather than measured as a variable of interest. As a result, little is known about the baseline first-fixation behavior of native RTL speakers in concurrent two-stimulus paradigms. Afsari et al. (2016) showed that reading direction primes can shift the spatial bias in bilingual RTL speakers, but the baseline behavior in the absence of explicit language priming has received less attention.

The findings of Ehinger et al. (2026) raise a further question: if RTL speakers have a reduced or reversed leftward bias, their data from a standard horizontal arrangement may already be relatively unbiased, whereas LTR speakers' data from the same arrangement would suffer substantial power loss. Characterizing the bias across reading-direction groups would therefore inform stimulus placement decisions in studies that test mixed-language samples.

1.2 Research Questions and Hypotheses

We ask whether native RTL speakers (Arabic and Persian) differ from native LTR speakers in their first-fixation behavior when viewing pairs of visual stimuli in a horizontal arrangement.

We expect LTR speakers to replicate the well-documented leftward preference. For RTL speakers, we predict a reduced leftward bias, consistent with line bisection and chimeric face studies. Whether the bias reverses entirely or merely diminishes is an open empirical question.

Dependent measures: first fixation proportion (the proportion of trials on which the participant fixated each side first) and first fixation reaction time (latency from stimulus onset to the first fixation on either stimulus).

Chapter 2

Experimental Methods

2.0.1 Participants

Eye tracking data were recorded from 23 participants; two were excluded from all analyses after quality control identified gaze calibration problems severe enough to make their data unusable (see Quality control below). The analyzed dataset includes the remaining 21 participants: 11 who selected a left to right (LTR) language (German or English) at the start of the session, and 10 who selected a right to left (RTL) language (Arabic or Persian). Within the LTR group, 9 participants chose English and 2 chose German; within the RTL group, 7 chose Persian and 3 chose Arabic. Group membership follows this self selected session language rather than a separate native language screening; participants were recruited as native speakers of the language they chose.

2.0.2 Materials and stimuli

Stimuli were 16 images (000.png through 015.png), each a 1025 x 1025 px PNG, organized into eight fixed pairs. Each pair was shown six times across the session, and its position in this cycle ran independently of the six trial block structure described below, so a given pair was not tied to any particular story segment.

On each trial, the two images of a pair appeared at the same time on opposite sides of a central fixation point, along one of six axis orientations spaced 30° apart (0°, 30°, 60°, 90°, 120°, 150° from horizontal). This follows the arrangement used by Ehinger et al. (2026). Within a block of six trials, each of the six orientations occurred exactly once, in an order randomized per participant.

To keep participants engaged with both images rather than only the one they looked at first, they were asked to count the circular shapes in the pair, distinguished from a similarly sized decagon shown once during the instructions. The count could run from zero to three per pair. After each trial the experimenter entered the spoken count by key press; this count was not analyzed as a dependent measure, it served only to keep participants looking at both stimuli rather than disengaging once one had been inspected.

Alongside the eye tracking task, participants read a children’s story (The Ugly Duckling) presented in nine segments. The first eight were shown one at a time, each before one six trial block; the ninth

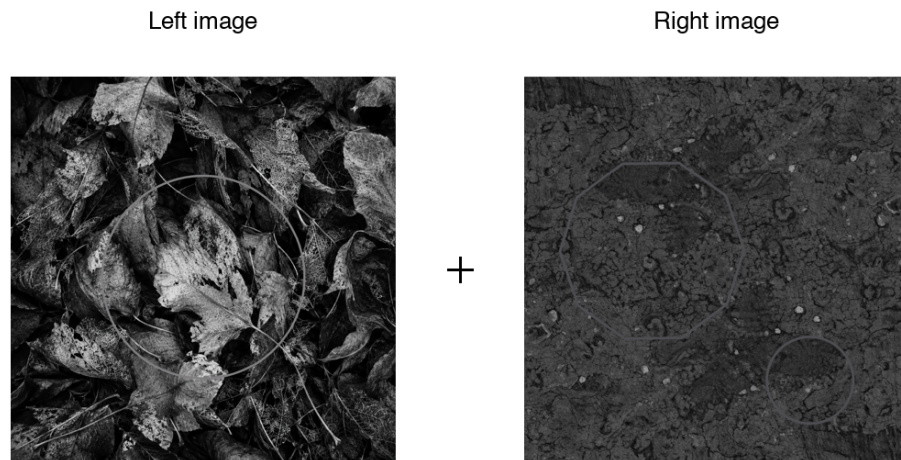


Figure 2.1: Example stimulus pair (pair 1 of 8) as presented on screen. Faint outline shapes embedded in the images are the circles participants were asked to count; the central cross marks the fixation point, not the on-screen fixation dot itself.

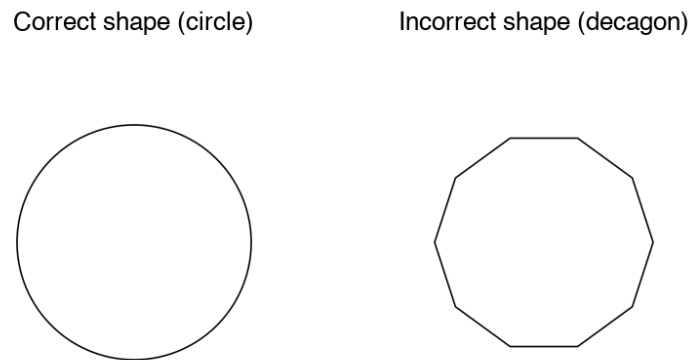


Figure 2.2: Shape distinction shown once during the instructions: the correct circle to count against the similarly sized incorrect decagon.

(the story’s resolution) was shown once after the final trial, followed by five comprehension questions asked aloud by the experimenter and recorded manually. The story gave participants sustained exposure to text in their session language and a natural framing for the task; the comprehension questions served as an attention check rather than a scored outcome.

2.0.3 Study design

The experiment used a between subjects design with reading direction group (LTR vs RTL) as the main factor. Each participant completed 48 trials in 8 blocks of 6, with one story segment shown at the start of each block. Within a block, the six trials sampled the six axis orientations in random order, so every participant saw every arrangement exactly 8 times over the session and arrangement is fully crossed with group rather than manipulated between subjects.

2.0.4 Experiment (implementation)

The task was built in OpenSesame 4.1.14 (“Neonatal Nightingale”) and run with a GazePoint GP3 HD eye tracker via PyGaze. The tracker ran at 150 Hz, one sample every 6.7 ms, and the recordings confirm it: the median interval between samples is 6.7 ms for every participant. Participants sat about 60 cm from a 1920 x 1080 px display with a visible area of 59.5 x 33.5 cm, which works out to 32.3 px per cm.

One detail of the display setup matters for the analysis. OpenSesame drew the experiment in a coordinate space of 1536 x 864, which Windows scaled by 125% to fill the 1920 x 1080 panel. Stimulus positions in the trial logs are therefore in the smaller space, while the eye tracker reports gaze relative to the physical display, and converting between the two is a step the preprocessing has to get right (see Preprocessing below).

Before the main task, participants chose their session language from four options (Deutsch, English, Arabic, Farsi) with a number key; this choice set both the group assignment and the language of the story text and on screen instructions.

The two images making up a pair were drawn from the same 1025 x 1025 px source files, shown at a fixed 0.25 scale. In the script’s own coordinate space that makes each image 256 x 256 px, drawn 308 px from the center of the screen; on the display, which renders that space 25% larger, the images were 320 x 320 px with their centers 770 px apart. In physical terms that is 9.9 cm per image with 23.9 cm between the centers, or about 9.4 degrees of visual angle per image and 22.5 degrees between the two at a viewing distance of 60 cm, far enough apart that the areas of interest described below could never overlap.

Each trial opened with a central fixation check: the script polled the tracker for up to 1.5 s for a gaze sample within 100 px of screen center, then continued regardless of whether fixation was reached. A fixed 1000 ms delay followed before the image pair appeared. Once the pair was on screen, the tracker was polled for 5000 ms; a gaze sample within 150 px of either image’s center on both the horizontal and vertical axis was logged online as the first fixation, recording its side, latency, and coordinates, but the pair stayed visible for the full 5 s regardless of when a fixation was detected. This online check tests individual gaze samples for proximity to a stimulus and does not run any fixation detection of its own; PyGaze’s event detection parameters were left at their defaults (35°/s and 9500°/s² for saccade velocity and acceleration) and play no part in what was

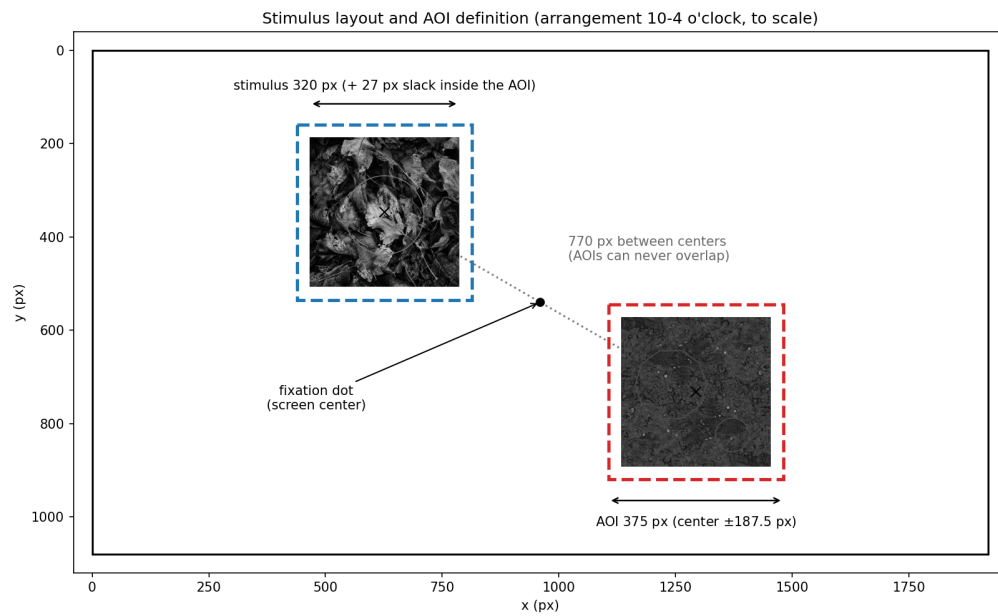


Figure 2.3: Stimulus layout and area-of-interest (AOI) definition for one arrangement (10-4 o'clock), drawn to scale. The two images are always equidistant from central fixation, along one of six axis orientations spaced 30° apart; a square margin of 187.5 px around each image center defines the AOI used for first-fixation detection. That is the 150 px margin the experiment script uses for its own online check, expressed in display pixels.

logged. The analysis reported here does not rely on the online values at all, and works instead from the fixation events the GazePoint tracker itself reports, as described under Preprocessing. Calibration was performed once at the start of the session; PyGaze logged accuracy (mean offset from the calibration targets, in degrees and pixels, separately for each eye) and precision (root mean square sample to sample noise) automatically for every participant.

2.1 Analysis methods

2.1.1 Preprocessing

The first fixation on each trial was identified from the raw gaze stream rather than taken from the value logged online during the session, so that the same detection rule could be applied uniformly and independently of what the OpenSesame script happened to record live. GazePoint’s own fixation classifier groups consecutive gaze samples into fixation events (identified by an FPOGID index, with normalized screen coordinates and a start time for each); these events were segmented into trials using the `STIMULUS_ONSET` and `IMAGE_OFFSET` markers written to the gaze log for that purpose. A fixation already in progress at stimulus onset was dropped before searching for the response, since it belongs to the preceding central fixation dot rather than to the stimuli that just appeared. Within the remaining window, the first fixation landing inside a square area of interest around either stimulus, on both axes, was taken as the trial’s response, recording its side and latency. The area of interest is the experiment’s own 150 px margin converted to display pixels, 187.5 px, applied here to tracker classified fixations rather than to individual raw samples.

That conversion is the part worth spelling out. The tracker reports gaze as a fraction of the physical display, so the stimulus coordinates in the trial logs, which live in OpenSesame’s 1536 x 864 space, have to be multiplied by 1.25 before the two can be compared. An earlier version of the pipeline added them straight to the center of a 1920 x 1080 frame, which left every area of interest about 77 px too close to the center of the screen. The fixation data say plainly which version is right: fixations made during a trial sit a median of 391 px from the center of the screen, against 385 px predicted by the scaled coordinates and 308 px predicted by the unscaled ones. Correcting it also improved agreement between the offline recomputation and the values the script logged live, from 90.4% to 92.2% of trials. The bias estimates moved very little either way, and no conclusion in this report depends on the correction, but the numbers reported here are the corrected ones throughout.

A related problem affected two subjects. Subjects 43 and 44 were recorded before the same display scaling issue was fixed in the experiment script itself, which left their online-logged first fixation side unusable: it registered a leftward or absent first fixation on all 96 of their combined trials, whereas the offline recomputation from the raw gaze stream found genuine rightward fixations for both of them. The offline recomputation does not depend on this part of the script and was used for every subject, not only these two, so the analysis pipeline is identical across the sample regardless of when a given participant was tested.

Trials were excluded when no fixation landed in either area of interest, or when the first fixation started less than 80 ms after onset, since a saccade that fast is planned before the stimulus appears rather than in response to it. No upper latency cutoff was applied. Slow first fixations were not random: they concentrated in arrangements without a stimulus in the upper left, so removing them would have removed a real part of the effect rather than noise; this pattern is examined on its own

terms in the latency analysis below rather than treated as measurement error.

In the raw log, `img_left` and `img_right` label the stimulus on the seven-to-twelve o'clock side of an arrangement and the one on the one-to-six o'clock side, not the physical left and right of the screen; for the vertical arrangement in particular, neither stimulus sits on the horizontal meridian, so calling one of them “the right stimulus” would really mean “the bottom stimulus.” To avoid this mislabeling, a trial’s outcome was instead recoded as the clock hour fixated first, since the six arrangements are simply the six diameters of a clock face. Left/right and up/down bias were then derived from this hour, each skipping the one arrangement lying on the axis perpendicular to it (12 to 6 for left/right, 9 to 3 for up/down), since those two positions sit on the meridian itself and carry no information for that contrast.

2.1.2 Sensitivity check: raw samples vs. classified fixations

The preprocessing above uses FPOGX/FPOGY, GazePoint’s own fixation classified gaze coordinates: consecutive raw samples are grouped into discrete fixation events (FPOGID) before the area of interest (AOI) test is applied. That classifier smooths and slightly delays the raw signal, so it is worth checking whether the same detection rule, applied to the unclassified gaze stream instead, changes the result. GazePoint also logs BPOGX/BPOGY (“Best Point of Gaze”), its estimate of gaze position for each raw sample, before any fixation grouping. The two are compared here directly, using the same AOI, trial window, and 80 ms anticipatory cutoff for both; the only thing that changes is whether “first fixation” means the first classified fixation event inside the AOI (FPOG) or the first raw sample inside it (BPOG):

```
def compute_bpog_first_gaze(subj, sub_trials):
    # Same AOI test as compute_fpog_first_fixation, but walks raw
    # per-sample BPOGX/BPOGY instead of grouping into FPOGID fixations.
    tsv = pd.read_csv(f"{DATA_DIR}/{subj}.tsv", sep="\t")
    tsv.replace(["NA", "undefined"], np.nan, inplace=True)
    for c in ["TIME", "BPOGX", "BPOGY", "BPOGV"]:
        tsv[c] = pd.to_numeric(tsv[c], errors="coerce")
    tsv = tsv.dropna(subset=["BPOGX", "BPOGY"]).sort_values("TIME").reset_index(drop=True)

    onsets = tsv[tsv["USER"].astype(str).str.contains("STIMULUS_ONSET", na=False)]
    offsets = tsv[tsv["USER"].astype(str).str.contains("IMAGE_OFFSET", na=False)]["TIME"].values

    rows = []
    for (_, onset_row), (_, trial) in zip(onsets.iterrows(), sub_trials.iterrows()):
        t0 = onset_row["TIME"]
        later = offsets[offsets > t0]
        t_end = later[0] if len(later) else t0 + 10.0

        window = tsv[(tsv["TIME"] >= t0) & (tsv["TIME"] <= t_end) & (tsv["BPOGV"] == 1) &
                      tsv["BPOGX"].between(0, 1) & tsv["BPOGY"].between(0, 1)]

        # DISPLAY_SCALE = 1.25: logged coordinates are in OpenSesame's
        # 1536x864 space, gaze is reported relative to the 1920x1080 display
```

```

lx = SCREEN_W / 2 + trial["x_left"] * DISPLAY_SCALE
ly = SCREEN_H / 2 + trial["y_left"] * DISPLAY_SCALE
rx = SCREEN_W / 2 + trial["x_right"] * DISPLAY_SCALE
ry = SCREEN_H / 2 + trial["y_right"] * DISPLAY_SCALE

side, rt_ms = "none", np.nan
for _, s in window.iterrows():
    xi, yi = s["BPOGX"] * SCREEN_W, s["BPOGY"] * SCREEN_H
    in_left = abs(xi - lx) < AOI_MARGIN_PX and abs(yi - ly) < AOI_MARGIN_PX
    in_right = abs(xi - rx) < AOI_MARGIN_PX and abs(yi - ry) < AOI_MARGIN_PX
    if in_left or in_right:
        side = "left" if in_left else "right"
        rt_ms = (s["TIME"] - t0) * 1000.0
        break
    rows.append(dict(subject_nr=str(trial["subject_nr"]), trial_nr=int(trial["trial_nr"]),
                    bpog_first_fix_direction=side, bpog_first_fix_rt_ms=rt_ms))
return pd.DataFrame(rows)

```

The resulting `bpog_first_fix_direction` was merged onto the same trial table, passed through the same `apply_exclusions()` used for FPOG (an 80 ms anticipatory cutoff, no upper cutoff), and compared across groups with the same hierarchical bootstrap used for the FPOG left/right and up/down proportions:

```

trials = apply_exclusions(trials, side_col="bpog_first_fix_direction",
                        rt_col="bpog_first_fix_rt_ms",
                        min_latency_ms=MIN_LATENCY_MS,
                        reason_col="bpog_exclusion_reason", excluded_col="bpog_excluded")

df_valid_bpog = trials[~trials["bpog_excluded"]].copy()
df_valid_bpog["physical_side"] = df_valid_bpog["bpog_fixated_clock"].map(hour_side)
df_valid_bpog["physical_vert"] = df_valid_bpog["bpog_fixated_clock"].map(hour_vert)

group_lr_bpog, _ = pooled_orientation_table(df_valid_bpog, "physical_side", "right", "left")
group_ud_bpog, _ = pooled_orientation_table(df_valid_bpog, "physical_vert", "up", "down")

```

At the trial level, the two measures agreed on direction for 93.9% of trials. The disagreements were lopsided: BPOG called “right” where FPOG called “left” more than three times as often as the reverse (47 vs. 14 trials). That fits a saccade briefly overshooting or sweeping past one AOI on its way to a stable fixation elsewhere. GazePoint’s fixation classifier correctly discards a sample like that, but a test that checks every raw sample cannot tell it apart from a genuine fixation. BPOG’s exclusion rate under the 80 ms anticipatory cutoff was also about five times higher than FPOG’s (5.7% vs. 1.1%), since a single raw sample can drift into an AOI for a few milliseconds in a way a fixation, which persists by definition, mostly cannot.

At the group level, the two measures agreed closely for LTR (largest difference across left/right and up/down: 0.041) and less closely for RTL (0.004 for left/right, 0.060 for up/down, where the FPOG estimate of 83.4% dropped to 77.4% under BPOG). RTL’s larger movement reflects its smaller

sample, 10 subjects against LTR’s 11: the same handful of relabelled trials shifts a 10-subject mean more than an 11-subject one, and it falls on the same arrangements (12-6, 7-1, 8-2) that already produced the slowest, noisiest first fixations under FPOG. Still, the direction of every effect held: RTL fixated right and up more often than LTR under both measures, and RTL’s upward bias, though smaller under BPOG, stayed clearly above chance. Recomputing first-fixation direction from raw samples adds noise, mostly through brief AOI crossings that are not real fixations, but it does not change any conclusion in this report. FPOG remains the measure used throughout the rest of the analysis.

2.1.3 Quality control

Before analysis, each recording was checked for complete trial markers (48 stimulus onsets, offsets, and fixation dot events), image display duration close to the intended 5 s, and effective sampling rate close to the tracker’s 150 Hz with few gaps longer than 50 ms. All 23 recordings passed on those counts: every one has its 48 markers, a median display duration within 5 ms of the intended 5 s, and a median sampling interval of 6.7 ms. Calibration quality was estimated per trial from gaze recorded during the central fixation dot: accuracy as the mean distance from screen center, and precision as the RMS spread around each trial’s own mean. These were compared across subjects with boxplots and an x/y offset scatter to catch systematic drift, and a small number of trials per subject were plotted as full scanpaths to confirm that gaze actually landed inside the areas of interest and that the arrangement positions matched the intended layout.

Two participants, 61 and 107, were excluded from all analyses for poor gaze quality. Participant 61’s gaze during the central fixation dot sat a median of 485 px from screen center, about 50% further off than any other participant’s and nearly constant across the session (interquartile range 8 px) rather than varying trial to trial as tracking noise would, and gaze reached one of the two stimuli on only 20 of 48 trials, against 44 or more for everyone else. Participant 107’s offset was smaller at 274 px, but it had the largest downward component in the sample by more than a factor of two, and their first fixations went to the lower stimulus on 87% of trials, which no other participant comes close to. Neither recording supports a first-fixation measure, so both were dropped before analysis. Among the 21 participants kept, calibration offset and up/down fixation bias are unrelated ($r = -0.07$), so drift does not explain the upward bias reported in the results.

2.1.4 Analysis pipeline

Preprocessing and analysis were carried out in Python 3.11, using pandas and numpy for data handling, scipy and statsmodels for the statistical tests, and matplotlib for the figures.

Group comparisons used a hierarchical bootstrap: subjects were resampled first, then trials within each resampled subject, since trials from the same subject are not independent and treating them as such would understate the true uncertainty. The point estimate is the unweighted mean of subject means, so that a subject with more valid trials does not automatically carry more weight; the confidence interval is the 2.5th and 97.5th percentile of 5,000 such resamples, drawn with a fixed random seed so the reported intervals are exactly reproducible from the same data. This procedure was applied to the left/right and up/down fixation proportions per group. The twelve hour clock-face profile is reported as the mean of subject means without intervals, since each hour rests on roughly eight trials per subject.

A logistic regression in the style of a Bradley-Terry model estimated the log odds of fixating the clockwise-numbered stimulus first as a function of arrangement, entered as a six level categorical predictor, and reading direction group, giving a single test of whether the two groups differ once arrangement is accounted for rather than six separate group comparisons. Standard errors are clustered by subject, for the same reason the group proportions above use a hierarchical bootstrap rather than a plain proportion: trials from the same subject are not independent. Arrangement and group enter as additive main effects only. Their interaction, which would give each arrangement its own group comparison, does converge, but with only 10 RTL subjects spread across twelve arrangement-by-group cells (one of which holds 2 of 79 trials), and no clustered version of that model fit, its per-arrangement estimates are not reliable enough to report. The additive model does not have this problem and is the one reported.

Because the outcome in that model is “fixated the one-to-six o’clock member of the pair,” it mixes the two axes: in the 9-3 arrangement that member is the right-hand stimulus, in 12-6 it is the bottom one. The same model was therefore also fitted to each axis separately, using the clock hour recoding described above: once to the left/right outcome with the 12-6 arrangement dropped, once to the up/down outcome with 9-3 dropped, and once to the 9-3 arrangement on its own, which is the only arrangement whose contrast is purely horizontal. All three use the same clustered standard errors.

First-fixation latency was examined separately by arrangement and group, since the spatial bias also showed up as a difference in response speed rather than only in which side was chosen. The latency distribution is bimodal, with a fast mode around 250 to 350 ms and a slow tail extending past 2 s; because no upper cutoff was applied during preprocessing, this tail is available for analysis rather than discarded. The median latency by arrangement and group changes little, since the fast mode dominates almost everywhere, so the two groups were also compared on the share of trials falling in the slow tail, using 2 s (the approximate valley between the two modes) as a descriptive boundary rather than an exclusion criterion.

Chapter 3

Results

The 21 participants retained after exclusions (Methods) contributed 997 valid first fixations out of 1008 trials: 11 LTR participants (523 trials) and 10 RTL participants (474 trials). The 11 dropped trials split into 7 first fixations faster than 80 ms and 4 trials with no fixation in either area of interest. All results below use the first fixation computed from FPOG, the tracker classified measure used throughout preprocessing. A sensitivity check against raw gaze samples instead of classified fixations, reported in Methods, leaves every conclusion here unchanged.

3.0.1 Left/right and up/down bias by group

Figure 3.1 and Table 3.1 give the pooled proportion of trials on which each group's first fixation went left, right, up, or down, with hierarchical bootstrap confidence intervals. `physical_side` and `physical_vert` are the same clock-hour-derived columns from Preprocessing, each skipping the one arrangement that carries no information for that axis:

```
def pooled_orientation_table(df, axis_col, positive_label, negative_label):
    rows, subj_rows = [], []
    for group, g in df.dropna(subset=[axis_col]).groupby("group"):
        per_subj = [(subj, (sg[axis_col] == positive_label).astype(float).values)
                     for subj, sg in g.groupby("subject_nr")]
        arrays = [a for _, a in per_subj]
        p_pos, lo_pos, hi_pos = hierarchical_bootstrap_ci(arrays)
        rows.append(dict(group=group, direction=positive_label, n_subjects=len(arrays),
                         n_trials=len(g), p=p_pos, ci_lo=lo_pos, ci_hi=hi_pos))
        rows.append(dict(group=group, direction=negative_label, n_subjects=len(arrays),
                         n_trials=len(g), p=1 - p_pos, ci_lo=1 - hi_pos, ci_hi=1 - lo_pos))
    for subj, a in per_subj:
        subj_rows.append(dict(group=group, subject_nr=subj,
                             direction=positive_label, p=a.mean()))
    return pd.DataFrame(rows), pd.DataFrame(subj_rows)
```

```
group_lr, subj_lr = pooled_orientation_table(df_valid_fpog, "physical_side", "right", "left")
group_ud, subj_ud = pooled_orientation_table(df_valid_fpog, "physical_vert", "up", "down")
group_dir = pd.concat([group_lr, group_ud], ignore_index=True)
```

Table 3.1: Pooled first-fixation direction by group (FPOG-based), with hierarchical bootstrap 95% confidence intervals.

Group	Direction	P(first fixation)	95% CI
LTR	left	71.8%	[64.2%, 79.5%]
LTR	right	28.2%	[20.5%, 35.8%]
LTR	up	72.6%	[62.5%, 81.5%]
LTR	down	27.4%	[18.5%, 37.5%]
RTL	left	49.0%	[37.4%, 60.5%]
RTL	right	51.0%	[39.5%, 62.6%]
RTL	up	83.4%	[76.9%, 89.3%]
RTL	down	16.6%	[10.7%, 23.1%]

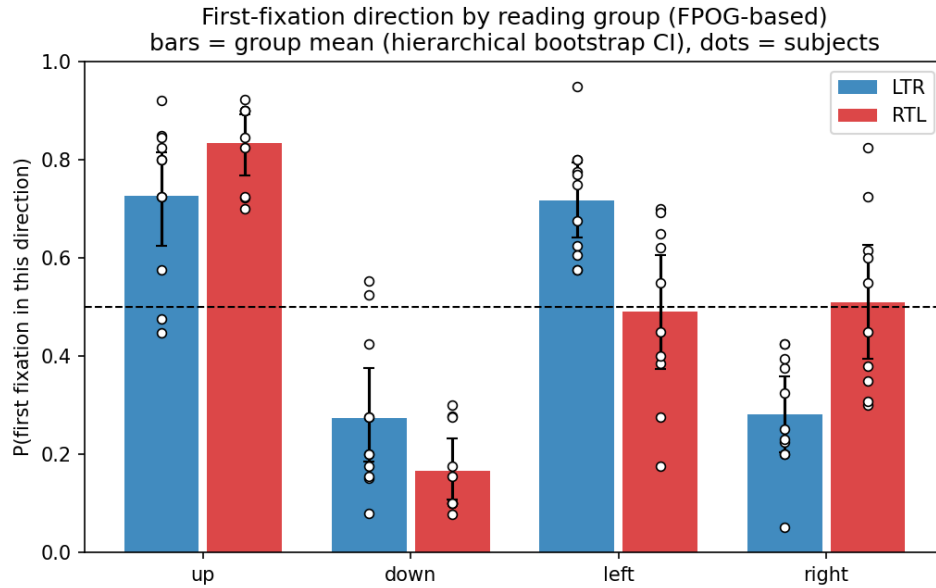


Figure 3.1: First-fixation direction by reading group. Bars are the group mean (hierarchical bootstrap CI); dots are individual subject means; the dashed line marks chance.

Both groups fixate up well above chance. On left/right the two groups part company. LTR's leftward bias of 71.8% sits inside the 60% to 93% range reported for comparable paradigms in the literature reviewed in the Introduction, and its interval is comfortably clear of chance. RTL's proportions land

almost exactly on 50%, with a marginal lean to the right (51.0%) and an interval, [39.5%, 62.6%], that spans chance with room to spare. In other words the RTL group as a whole has no measurable horizontal preference on this measure, which is the direction the hypotheses predicted but a weaker result than a reversal would be. The two intervals overlap, so the raw proportions alone cannot settle whether the groups differ; the regression below is the test that can.

The up/down contrast runs the other way. RTL's upward bias of 83.4% is larger than LTR's 72.6%, and RTL's interval is the tightest of the four. Nothing in the hypotheses anticipated a group difference on the vertical axis.

3.0.2 Clock-face profile

Figure 3.2 breaks the same bias down by all twelve clock positions rather than collapsing to left/right and up/down. Each trial contributes to the two hours actually presented in its arrangement, so every hour's estimate is conditioned on having been on screen at all:

```
rows = []
for _, r in df_valid_fpog.dropna(subset=["fixated_clock"]).iterrows():
    for h in r["arrangement"]:
        rows.append(dict(group=r["group"], subject_nr=r["subject_nr"], hour=int(h),
                        fixated=float(r["fixated_clock"] == h)))
clock_long = pd.DataFrame(rows)

subj_clock = (clock_long.groupby(["group", "subject_nr", "hour"])["fixated"]
              .agg(["mean", "count"]).reset_index())
group_clock = (subj_clock.groupby(["group", "hour"])["mean"]
               .mean().reset_index(name="p")) # mean of subject means
```

Both groups favour the top of the clock and avoid the bottom. Every hour from 10 to 1 sits above chance in both groups, and no hour from 4 to 7 does in either. The upper half is where the two profiles separate. LTR peaks on the left flank, at 11 o'clock (91.7%) and 9 o'clock (87.5%), while RTL peaks at 12 (97.5%) and spills to the right, with hours 1 and 2 at 83.4% and 79.5%. The pair of hours on the horizontal meridian makes the same point most directly, since neither carries any vertical component: LTR fixates 9 o'clock first on 87.5% of trials and 3 o'clock on 12.5%, whereas RTL splits them 51.2% to 48.8%.

Two individual hours diverge more than the pooled contrasts suggest, although each hour rests on about eight trials per subject and is correspondingly noisier. Hour 2, in the upper right, is RTL's largest lead anywhere on the clock: 79.5% against LTR's 35.2%. Hour 6, at the bottom, is the most extreme point on the plot for either group. LTR still fixates it first on 13.6% of trials, against 2 of 79 trials for RTL. Because opposite hours are two views of the same choice, the mirrored hours (8 and 12) tell the same story rather than adding to it.

3.0.3 Individual-subject bias vectors

Figure 3.3 plots each subject as a single point: horizontal bias ($P(\text{right}) - P(\text{left})$) against vertical bias ($P(\text{up}) - P(\text{down})$), each rescaled from a 0 to 1 proportion to a -1 to 1 bias so that 0 means no preference:

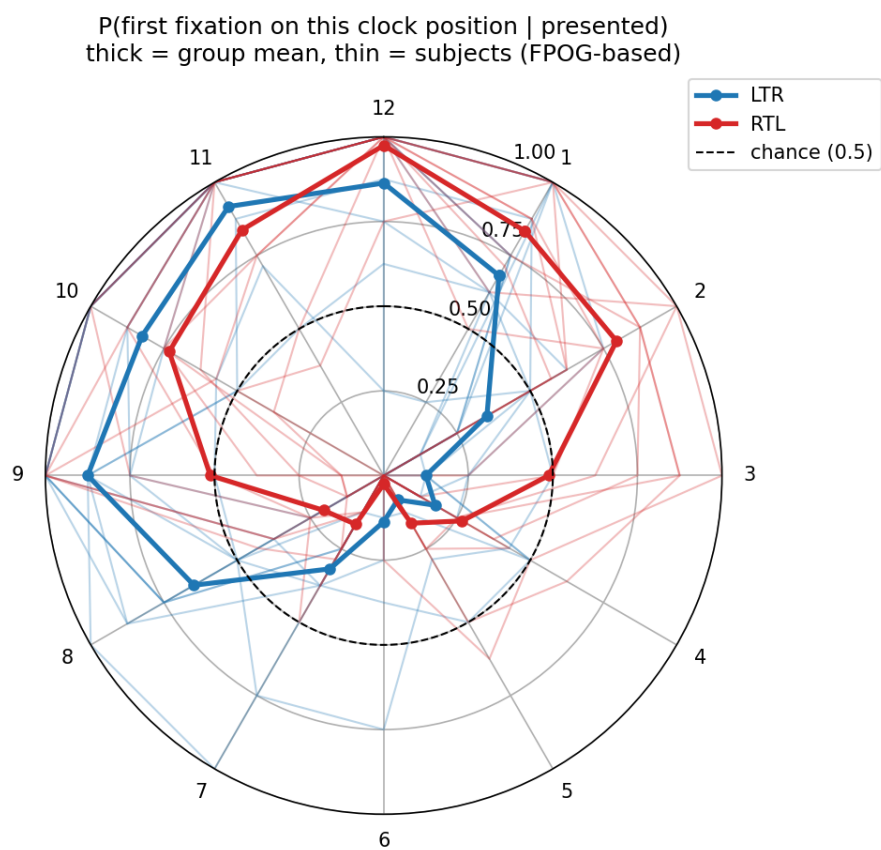


Figure 3.2: P(first fixation on this clock position, given it was presented), by group. Thick lines are group means; thin lines are individual subjects.

```

subj_rows = []
for (group, subj), g in df_valid_fpog.groupby(["group", "subject_nr"]):
    px = (g["physical_side"].dropna() == "right").astype(float).values
    py = (g["physical_vert"].dropna() == "up").astype(float).values
    subj_rows.append(dict(group=group, subject_nr=subj,
                          x=2 * px.mean() - 1, y=2 * py.mean() - 1))
quad_subj = pd.DataFrame(subj_rows)

group_rows = []
for group, g in df_valid_fpog.groupby("group"):
    side_arrays = [(sg["physical_side"].dropna() == "right").astype(float).values
                   for _, sg in g.groupby("subject_nr")]
    vert_arrays = [(sg["physical_vert"].dropna() == "up").astype(float).values
                   for _, sg in g.groupby("subject_nr")]
    px, px_lo, px_hi = hierarchical_bootstrap_ci(side_arrays)
    py, py_lo, py_hi = hierarchical_bootstrap_ci(vert_arrays)
    group_rows.append(dict(group=group, x=2 * px - 1, y=2 * py - 1,
                          x_ci_lo=2 * px_lo - 1, x_ci_hi=2 * px_hi - 1,
                          y_ci_lo=2 * py_lo - 1, y_ci_hi=2 * py_hi - 1))
quad_group = pd.DataFrame(group_rows)

```

The LTR mean falls in the up-left quadrant ($x = -0.44$, $y = 0.45$) with both intervals clear of zero, so LTR is reliably biased up and to the left as a group. The RTL mean sits just across the vertical axis at $x = 0.02$, $y = 0.67$: reliably upward, since its vertical interval excludes zero, but horizontally indistinguishable from no preference, with an interval of $[-0.21, 0.26]$ straddling zero. That matches Table 3.1.

The individual points are worth reading alongside the means. Five subjects lean rightward, and all five are RTL: subjects 54 ($x = 0.10$), 65 (0.20), 111 (0.23), 64 (0.45) and 66 (0.65), the largest rightward lean in the sample. No LTR subject does; the most extreme in that group is subject 108 at $x = -0.90$. On the vertical axis the picture is more uniform. Every RTL subject has a positive value, between 0.40 and 0.85, and only two participants anywhere in the sample fall below zero: LTR subjects 101 (-0.05) and 106 (-0.11), both close enough to zero to read as no vertical preference rather than a downward one.

3.0.4 A single test of the group difference

The proportions above answer four separate questions (left/right and up/down, for each group) rather than the one question of primary interest: do LTR and RTL differ once each arrangement's own baseline preference is accounted for? Methods describes a logistic regression built for exactly this: the log odds of fixating the clockwise-numbered stimulus in each pair (`img_right`, the one-to-six o'clock member), as a function of arrangement (six levels) and group:

```

import statsmodels.formula.api as smf

df_bt = df_valid_fpog.copy()
df_bt["y_right"] = (df_bt["fpog_first_fix_direction"] == "right").astype(int)

```

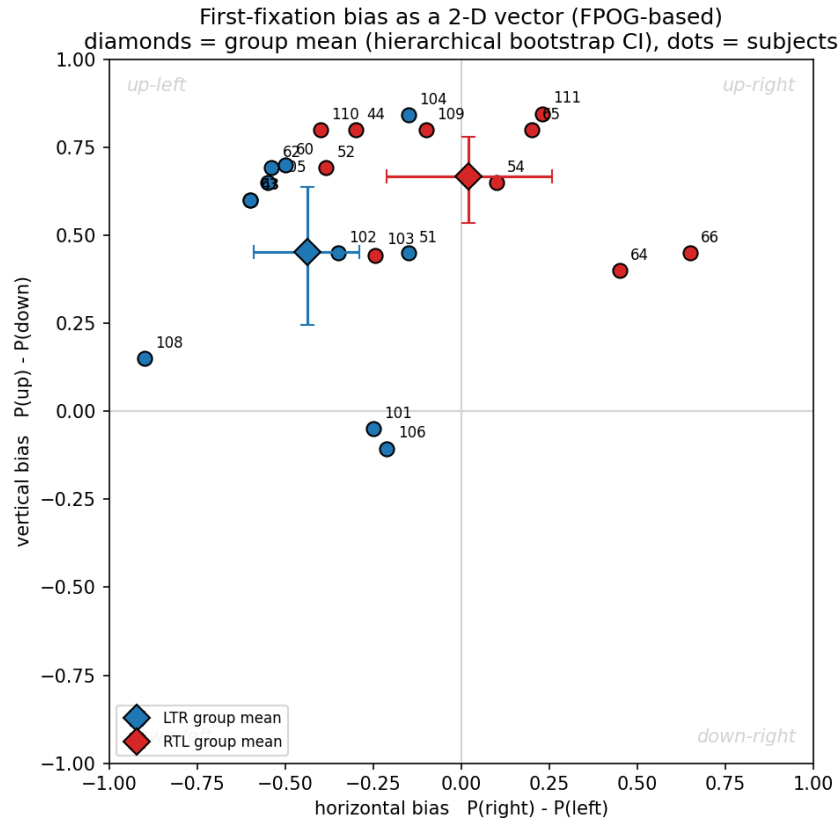


Figure 3.3: First-fixation bias as a 2-D vector. Diamonds are group means with hierarchical bootstrap CIs; dots are individual subjects.

```
df_bt["arrangement_str"] = df_bt["arrangement"].astype(str)
```

```
bt_model = smf.logit("y_right ~ C(arrangement_str) + C(group)", data=df_bt).fit(
    disp=0, cov_type="cluster", cov_kws={"groups": df_bt["subject_nr"]})
```

The model converges cleanly ($n = 997$) and standard errors are clustered by subject, since trials from the same subject are not independent, the same reason the proportions above use a hierarchical bootstrap.

Arrangement drives most of the fit. Relative to the baseline arrangement, 10 and 4 o'clock, the two arrangements without an upper-left stimulus (7 and 1 o'clock; 8 and 2 o'clock) sharply raise the odds of fixating the clockwise-numbered member of the pair ($b = 2.54$, $p < .001$; $b = 1.63$, $p < .001$). Those are also two of the three arrangements with the largest slow latency tails below. The remaining two non-baseline arrangements (11 and 5 o'clock; 12 and 6 o'clock) lower the odds instead ($b = -0.74$, $p = .005$; $b = -1.15$, $p = .022$), and only 9 and 3 o'clock does not differ from baseline ($p = .173$).

Group is significant in the predicted direction: the RTL coefficient is $b = 1.08$ (cluster SE = 0.41, $p = .009$). Fit without clustering the same coefficient reaches $p < .001$, which overstates the case: that version treats all 997 trials as independent observations when there are only 21 subjects behind them, so it is not the one reported. The clustered result is the one that matters, and it says the groups differ once each arrangement's own pull is taken out, even though RTL's raw left/right proportion on its own sits at chance.

An arrangement by group interaction, which would give each arrangement its own group comparison instead of one pooled test, was checked and not used:

```
inter_model = smf.logit("y_right ~ C(arrangement_str) * C(group)", data=df_bt).fit(disp=0)
# converges, but in the (12, 6)/RTL cell only 2 of 79 trials went to
# the clockwise member, which leaves that comparison on very little
```

Every cell is populated, so the model converges, but with 10 RTL subjects spread across twelve arrangement-by-group cells and no clustered version of it fitted, its per-arrangement estimates are not solid enough to report next to the additive model.

3.0.5 The same test, one axis at a time

The outcome in that model, "fixated the one-to-six o'clock member," is a horizontal contrast in the 9-3 arrangement and a vertical one in 12-6, so the group coefficient mixes the reading-direction effect with the vertical difference seen above. Refitting the same model to each axis separately keeps the two apart:

```
def group_effect(df, outcome_col, positive, label, with_arrangement=True):
    d = df.dropna(subset=[outcome_col]).copy()
    d["y"] = (d[outcome_col] == positive).astype(int)
    d["arrangement_str"] = d["arrangement"].astype(str)
    formula = "y ~ C(arrangement_str) + C(group)" if with_arrangement else "y ~ C(group)"
    return smf.logit(formula, data=d).fit(disp=0, cov_type="cluster",
                                         cov_kws={"groups": d["subject_nr"]})
```

```
horizontal_only = df_valid_fpog[df_valid_fpog["arrangement"].astype(str) == "(9, 3)"]
group_effect(df_valid_fpog, "physical_side", "right", ...) # drops 12-6
group_effect(df_valid_fpog, "physical_vert", "up", ...) # drops 9-3
group_effect(horizontal_only, "physical_side", "right", ..., with_arrangement=False)
```

Table 3.2: The group effect split by axis. All four models cluster standard errors by subject.

Outcome	Trials	b (RTL)	SE	p
clockwise member, all six arrangements	997	+1.08	0.41	.009
P(right), 12-6 dropped	832	+1.32	0.38	.001
P(up), 9-3 dropped	829	+0.70	0.34	.037
P(right) in 9-3 alone	168	+1.90	0.65	.004

Both axes carry a group difference, and the horizontal one is the stronger of the two ($b = 1.32$ against $b = 0.70$). The cleanest version of it is the last row: in the 9-3 arrangement, where the choice is purely left against right and no vertical preference can contribute, LTR fixates the right-hand stimulus first on 12.5% of trials and RTL on 48.8%. That is the reading-direction prediction in its most direct form, and it holds even though RTL's pooled left/right proportion across all arrangements sits at chance.

3.0.6 First-fixation latency by arrangement

Figure 3.4 shows the share of trials with a first fixation slower than 2000 ms, separately by arrangement and group; 2000 ms is a descriptive boundary at the valley of the bimodal latency distribution, not an exclusion criterion:

```
SLOW_MS = 2000

lat = df_valid_fpog.dropna(subset=["fpog_first_fix_rt_ms"]).copy()
lat["arrangement_lbl"] = lat["arrangement"].apply(lambda a: f"{a[0]}-{a[1]}")
lat["slow"] = lat["fpog_first_fix_rt_ms"] > SLOW_MS

med = lat.groupby(["group", "arrangement_lbl"])["fpog_first_fix_rt_ms"].median().unstack(0)
pslow = lat.groupby(["group", "arrangement_lbl"])["slow"].mean().unstack(0)
```

Median latency for both groups (**med** above) stays between 248 and 285 ms across every arrangement and is not shown here, since the fast mode dominates almost everywhere and the group difference lives in the slow tail (**pslow**) instead.

The slow tail concentrates in the three arrangements without an upper-left stimulus (7-1, 8-2, 12-6) for both groups, and in all three it is larger for RTL than for LTR: 7.6% against 5.7% in 7-1, 7.6% against 6.8% in 8-2, and 10.1% against 8.1% in 12-6. The other three arrangements offer an upper-left target and the tail nearly vanishes, reaching exactly 0% for both groups in 11-5 and for RTL in 10-4. So the same pattern that shows up in which stimulus was chosen also shows up in how

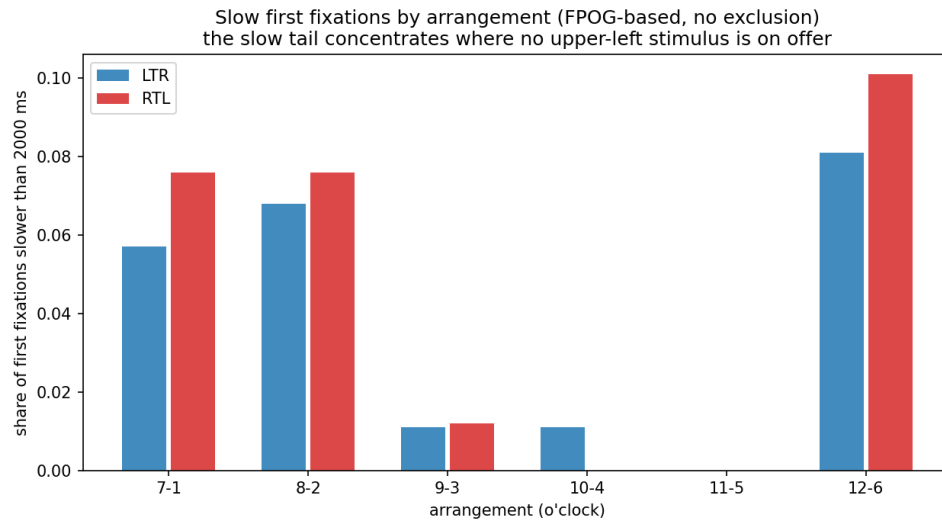


Figure 3.4: Share of first fixations slower than 2000 ms, by arrangement and group.

long the choice took: RTL is slowest precisely where its preferred upper-left target is missing. These are raw shares with no confidence intervals, and each RTL cell holds 78 to 80 trials, so they carry the same caveat about sample size as the proportions do.

Chapter 4

Discussion

LTR speakers replicated the well-documented leftward bias, and both groups fixated upward well above chance. RTL speakers' leftward proportion sat almost exactly at chance and, on its own, was indistinguishable from no preference at all. The confirmatory test tells a clearer story than that proportion does. Once each arrangement's own baseline pull is accounted for, the two groups differ ($b = 1.08$, $p = .009$), and splitting the test by axis puts most of that difference on the horizontal one ($b = 1.32$ for left/right against $b = 0.70$ for up/down). The single cleanest comparison is the 9-3 arrangement, where the choice is left against right with no vertical component at all: LTR looked right first on 12.5% of trials, RTL on 48.8%. Arrangement still explains more of the variance than group does, but the group effect is not marginal and it points where reading direction predicts.

4.0.1 Relation to the literature

LTR's leftward proportion (71.8%) sits inside the 60-93% range reported by Foulsham et al. (2013) and Ossandón et al. (2014), though below the roughly 80% that Ehinger et al. (2026) found for German speakers in the same horizontal arrangement. Composition probably accounts for part of that gap: our LTR group is nine English and two German speakers, not a purely German sample. The counting task may account for another part, since it asked participants to engage with both images rather than glance and move on. Neither would contradict the earlier result.

RTL's proportion (49.0%) fits a "reduced, not reversed" account. Chokron and Imbert (1993) and Rashidi-Ranjbar et al. (2014) found weaker or reversed leftward bias in Arabic, Hebrew, and mixed-reading-habit populations doing line bisection; ours lands on chance rather than reversing outright. Afsari et al. (2016) showed native RTL readers shift rightward after an RTL priming passage but leftward after an LTR one, while late-acquired RTL readers show no such flexibility. Our participants read a story in their own language but were not primed with the contrast, so a near-chance default is a sensible baseline for priming to move away from in either direction. That the pooled proportion sits at chance while the arrangement-adjusted regression finds a reliable difference suggests the horizontal habit is real but weak enough to be swamped, in a simple proportion, by each arrangement's own pull.

The vertical result is the one the hypotheses did not anticipate. RTL's upward bias (83.4%) is

larger than LTR’s (72.6%), and reading direction is a horizontal habit; neither the right-hemisphere lateralization account (Corbetta and Shulman 2002) nor the cross-linguistic account predicts a group difference on this axis. The arrangement effects in the regression suggest a possible reason. The three arrangements without an upper-left stimulus (7-1, 8-2 and 12-6) are exactly the three where RTL’s slow-latency tail grows, and 12-6 is where its vertical preference becomes near-absolute: RTL fixated the lower stimulus first on 2 of 79 trials, against 13.6% of trials for LTR. That reads as a fallback: when an arrangement offers no upper-left target for the horizontal bias to lock onto, both groups lean harder on the vertical one, and RTL leans hardest because it has less horizontal pull to begin with. The data are consistent with this account without confirming it. The arrangement-by-group interaction that would test it directly does converge, but with 10 RTL subjects spread across twelve cells and no clustered fit computed, it is not a reliable per-arrangement test (see Limitations). If the account is right, the upward bias would be closer to a general property of the attentional system than a culturally modulated one, a distinction the horizontal-axis literature reviewed in the Introduction does not address.

Individual variability sits underneath all of this. Foulsham et al. (2018) report that leftward viewing biases are stable within a person but unrelated to handedness or line-bisection performance, that is, a real trait that does not reduce cleanly to the usual explanations. Our quadrant plot shows the same texture. Five subjects lean rightward and no LTR subject is among them, which lines up with the group means, but the LTR points themselves range from -0.15 to -0.90 on the same axis, and two of them show no vertical preference at all while every RTL subject does. The group means are real, and they sit on top of individual variation that the group-level story does not reach.

4.0.2 Limitations

The RTL group is smaller than LTR’s, ten subjects and 474 trials against eleven and 523, and every RTL confidence interval reflects it. The descriptive left/right proportion is the clearest case: at 49.0% it cannot be told apart from no preference on its own, and only the arrangement-adjusted models recover the group difference. Clustering standard errors by subject, the correct choice given non-independent trials, is what separates the regression’s naive $p < .001$ from its clustered $p = .009$. The effect survives that correction on both axes, but with 21 subjects the intervals stay wide, and this is the same power problem Ehinger et al. (2026) describe for spatial-bias designs generally.

Group membership follows the native language a participant selected and was recruited under, rather than a formal history questionnaire covering age of acquisition or biliteracy, so a lateralization confound from atypical or mixed reading histories cannot be fully ruled out. Reading direction is also confounded with everything else that differs between four languages and their speakers: culture, script, the translated story text, the recruitment pool. A between-subjects design with 21 people cannot separate reading direction from these co-varying factors; it can only describe the group difference, not attribute it.

The arrangement-by-group interaction, which would give each arrangement its own group comparison and test the vertical fallback account directly, converges but is not trustworthy at this sample size: 10 RTL subjects across twelve cells, one of them nearly unanimous (2 of 79 trials went to the lower stimulus), and no clustered version of the model fitted. The hour-by-hour and arrangement-by-arrangement differences described above therefore remain descriptive.

The counting task required participants to engage with both images rather than only the one fixated

first, which is a deliberate departure from pure free viewing. First-fixation latency stayed fast, with medians between 248 and 285 ms in every arrangement, which argues against much top-down influence on the very first saccade, but it does not rule out a subtler effect of knowing a count would be required.

4.0.3 Challenges during the project

The clock-face recoding turned out to be necessary, not optional. The raw log’s `img_left/img_right` labels track each pair’s position along the clock diameter, not physical screen sides, so for every non-horizontal arrangement one of “left” and “right” was really “top” or “bottom”. This surfaced only once the vertical and diagonal arrangements were analyzed, and the fix, recoding every trial by the clock hour actually fixated, had to happen before any left/right or up/down contrast could be trusted.

Display scaling caused trouble twice, in two different places. Two participants (43, 44) were run before a scaling bug in the experiment script was caught, and it made their online-logged first-fixation values unusable: both showed a leftward or absent first fixation on all 96 combined trials, which offline recomputation from the raw gaze stream showed to be wrong in both cases. That recomputation became the pipeline for every participant once the online log could no longer be trusted uniformly. The same distinction between OpenSesame’s 1536 x 864 coordinate space and the 1920 x 1080 display then bit the offline pipeline too, which compared gaze in display pixels against stimulus positions in script coordinates and so placed every area of interest about 77 px too close to the center of the screen. Fixation positions during the trials made the error visible and easy to check, and correcting it left the conclusions intact, but it is a reminder that a coordinate space is worth writing down explicitly rather than inferring.

Recruiting native RTL speakers was harder than recruiting LTR speakers, which is the direct cause of the eleven-to-ten imbalance and, downstream, of the power problem described above. Quality control also caught calibration problems in two participants (61, 107) only after full data collection, when the sessions could no longer be corrected. And running the study in four languages (German, English, Arabic, Persian) meant translating the story text, instructions, and comprehension questions into all four, coordinated across a small team, on top of the core eye-tracking design.

4.0.4 Outlook

Adding a formal native-language history questionnaire, covering age of acquisition and biliteracy rather than self-reported language selection alone, would narrow the lateralization confound instead of leaving it as a design limitation.

The vertical-fallback account is testable on its own. A follow-up powered specifically for an arrangement (upper-left target present versus absent) by group interaction would confirm or rule it out, rather than relying on the descriptive pattern reported here. It also suggests a concrete revision to Ehinger et al.’s (2026) stimulus-placement recommendation for mixed-language samples: rather than picking one diagonal arrangement that is neutral for a single population, a design could place each group’s preferred target so both groups’ biases are equally engaged, since the arrangements that produced the largest group divergence here were exactly the ones offering neither group its usual upper-left anchor.

Two things would be worth changing about the pipeline itself. Logging the clock-hour position directly, instead of `img_left/img_right`, would remove the recoding step and the ambiguity that could have gone unnoticed if the vertical and diagonal arrangements had not been checked closely. Logging stimulus positions in display pixels, or at least recording the display scale alongside them, would have prevented both scaling problems outright. Running the offline recomputation against a few pilot sessions before full data collection would also have caught the first of them immediately, rather than after two participants' data needed a separate fix.

Chapter 5

Contribution table

Task	Junichi	Sanaz	Nikhil	Shehab
Background Literature	-	-	—	—
Experiment Design	-	-	—	—
Stimulus Design	—	—	-	-
Piloting	—	—	—	-
Data-Recording	—	-	-	—
Non-Final-Talk presenting (who talks)	—	-	—	-
Non-Final-Talk presenting (who prepares)	—	-	—	-
Final-Talk presenting (who talks)	—	—	—	-
Final-Talk presenting (who prepares)	—	-	-	-
Data Analysis Scripts	—	-	—	-
Report Writing	—	—	-	-
Report Lectorate	—	-	-	-

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